



CENG 408

**Innovative System Design and
Development 2**

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Project:

**Syntagma: AI Powered Spaced Repetition System
Based Language Learning App**

Team Members:

202211403 İlker Feza Çoban

202111027 İlayda Sümer

202111007 Elif Su Tufan

202211077 Alp Yılmaz

202211032 Sultan Gürbüz

202111417 Ömerhan Yiğit

Advisor: Assist. Prof. Dr. Abdül Kadir GÖRÜR

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1. Project Summary

Syntagma is an AI-powered language learning application designed to help Turkish-speaking learners acquire English vocabulary through spaced repetition, contextual learning, and AI-driven explanations. The project encompasses three major components: a Spring Boot backend REST API, a React Native mobile application, and a Chrome browser extension, all working together to deliver a cohesive, multi-platform learning experience.

The application targets a real gap in existing language learning tools: the lack of an integrated system that combines scientifically backed spaced repetition scheduling with in-context vocabulary acquisition from real-world content such as web pages, videos, and eBooks.

2. Key Achievements

2.1 FSRS-Based Spaced Repetition Engine

The core of Syntagma is its implementation of the Free Spaced Repetition Scheduler [1] (FSRS) algorithm, a modern, research-backed alternative to the classic SM-2 [2] algorithm. The backend SRS engine manages card states (stability, difficulty, reps, lapses), computes optimal review intervals, and serves daily review queues that mix due cards with new cards. The system automatically promotes words to "KNOWN" status after sustained recall success (≥ 3 successful reviews with a scheduled interval exceeding 25 days), ensuring that mastered vocabulary exits the review cycle organically.

2.2 AI-Powered Language Assistance

A dedicated AI service integrates with the OpenRouter API [3] to provide three distinct AI-powered features, all tailored for Turkish-speaking learners:

Word Explanation: Given a word and its surrounding sentence, the AI returns the Turkish meaning, part of speech, usage notes, common mistakes, and example sentences — all in Turkish.

Sentence Translation: Provides natural, literal, and alternative Turkish translations for English sentences, helping learners understand nuance and idiomatic usage.

Sentence Grammar Explanation: Breaks down sentence structure into grammatical chunks, explains tense/aspect/voice in Turkish, and offers concrete learner tips.

All AI calls are routed through the backend (/api/ai endpoints), keeping API keys secure and enabling centralized rate limiting and monitoring.

2.3 Chrome Extension Learning Features

The Syntagma Chrome extension (Manifest V3) is a comprehensive learning companion that operates directly within the learner's browsing environment. Key capabilities include:

In-Page Word Popup: Users can click or select any word on a web page to see its definition, AI explanation, and instantly create a flashcard — all without leaving the page.

EPUB eBook Reader: A built-in reader allows users to upload and read EPUB files within the extension. Words can be looked up and turned into flashcards in context, with reading progress tracked and synced to the backend.

Video Learning with Audio/Screenshot Capture: The extension captures sentence-level audio and screenshots from videos (e.g., YouTube subtitles), enabling multimedia flashcard creation with real audio pronunciations and visual context.

Page Comprehension Analysis: Analyzes page text against the user's known vocabulary to estimate comprehension percentage, helping learners choose appropriate reading material.

Known Words Management: Users can view, filter, and manage their known words list. CEFR-level intake allows bulk-marking words as known based on proficiency level (A1–C2).

Manual Flashcard Creator: A dedicated card creator page allows users to build flashcards manually with custom front/back content, audio, and images.

2.4 Mobile Application

The React Native (Expo) mobile app provides on-the-go access to the learning system with the following screens and features:

Flashcard Review: Interactive review sessions using the SRS scheduling grading and session summaries showing performance metrics.

Flashcard Library: Browse, search, and manage the complete flashcard collection with filtering by knowledge status.

Badge System: Achievement badges reward milestones (cards reviewed, streaks, words mastered), increasing user motivation and engagement.

Push Notifications: Daily review reminders via expo-notifications keep learners on track with their spaced repetition schedule.

Settings & Language Management: Users can configure daily card limits, manage target languages, and customize their learning experience.

2.5 Robust Backend Architecture

The Spring Boot backend, built with Java 17 and backed by PostgreSQL, provides a comprehensive REST API with the following architectural highlights:

JWT Authentication & Security: Token-based authentication with Spring Security, including email verification and secure password handling.

Cloud Storage Integration: S3-compatible object storage (DigitalOcean Spaces) for eBook files and media assets, with pre-signed URL generation for secure uploads and downloads.

Offline-First Synchronization: A dedicated sync service (SyncController/SyncService) enables both mobile and extension clients to push local changes and pull server updates, supporting offline usage with eventual consistency.

Collection Management: Flashcards can be organized into user-defined collections, with full CRUD operations and item management.

Media Asset Management: Audio recordings and screenshot images are stored as MediaAsset entities linked to flashcards, supporting rich multimedia cards.

Review Analytics: Review logs track every review event with timestamps, device info, and results, enabling detailed learning analytics and progress reporting.

OpenAPI Documentation: Springdoc integration provides auto-generated Swagger UI for API exploration and testing.

2.6 Word Knowledge Tracking

The WordKnowledge system maintains a per-user vocabulary profile with four statuses: KNOWN, LEARNING, UNKNOWN, and IGNORED. The system supports batch updates, CEFR-level intake (automatically marking all words at or below a given proficiency level as known), and bidirectional integration with the flashcard review cycle. Words that are mastered through spaced repetition are automatically promoted to KNOWN status.

3. Limitations

Single Language Pair: The current implementation is specifically designed for Turkish speakers learning English. The AI prompts, CEFR word lists, and UI text are hardcoded for this pair. Supporting additional language combinations would require significant refactoring of the AI prompt templates and vocabulary databases.

AI API Dependency: The AI features depend on the OpenRouter [3] API (external third-party service). This introduces latency, cost per request, and a single point of failure. There is no fallback mechanism if the AI service is unavailable.

Limited Automated Testing: While some unit tests exist (audio recorder, screenshot capture, flashcard logic, page analysis), the overall test coverage -particularly for integration tests and end-to-end flows - is limited. The backend uses H2 for test environments but lacks comprehensive API-level test suites.

Extension Platform Lock-In: The browser extension is built exclusively for Chrome (Manifest V3 [4]). Firefox, Safari, and other browsers are not supported.

Scalability Concerns: The current architecture uses a single PostgreSQL instance with no caching layer (e.g., Redis). Under high user load, the AI proxy endpoints and SRS computation could become bottlenecks.

4. Lessons Learned

AI Prompt Engineering Is Iterative: Designing AI prompts that consistently return valid JSON with Turkish-language content required multiple iterations. Early versions produced inconsistent

formatting, mixed-language responses, or hallucinated grammar terms. Structured JSON schemas in the system prompt and explicit language requirements significantly improved reliability.

Multi-Platform Development Increases Complexity Exponentially: Maintaining three client platforms (backend, mobile, extension) with shared data models but different runtime environments (JVM, React Native [5], Chrome extension sandbox) created significant coordination overhead. Schema changes required updates across all three codebases.

Chrome Extension Manifest V3 Constraints: Migrating to Manifest V3 introduced challenges with service worker lifecycle management (no persistent background pages), storage limitations, and content script injection timing. The service worker's ephemeral nature required careful state management using Chrome Storage API and IndexedDB.

Offline-First Is Harder Than Expected: Building reliable offline support with eventual sync required rethinking data flow. Edge cases — such as creating a flashcard offline, reviewing it before syncing, then syncing with a server that assigns different IDs — demanded careful handling of local vs. server identity.

5. Conclusion, Shortcomings, Future Works

Syntagma demonstrates that integrating spaced repetition science with modern AI capabilities and multi-platform delivery can create a powerful, context-aware language learning tool. By embedding vocabulary acquisition directly into the learner's natural browsing and reading activities rather than confining it to a standalone app, the system lowers the barrier to consistent practice.

The FSRS algorithm ensures that review scheduling is both efficient and personalized, while the AI-powered explanations provide the kind of contextual, native-language support that traditional flashcard apps lack. The combination of a Chrome extension for in-context learning, a mobile app for on-the-go review, and a robust backend for synchronization creates a genuinely multi-platform learning ecosystem.

While limitations remain particularly around language pair flexibility, test coverage, and scalability the foundation is solid and extensible. The lessons learned during development, especially around AI prompt engineering, and cross-platform data synchronization, provide valuable guidance for the future evolution of the platform. Syntagma stands as a functional prototype that validates the core thesis: AI-enhanced spaced repetition, delivered in context, can meaningfully improve the vocabulary acquisition process for language learners.

5.1. Backend Application

5.1.1 Conclusion

The Spring Boot backend was successfully implemented, providing JWT-based authentication, FSRS spaced repetition scheduling, flashcard and collection management, eBook storage via S3-compatible presigned URLs, word knowledge tracking, and AI-powered language assistance

through the OpenRouter API. The REST API serves both the Chrome extension and mobile application with a unified data model backed by PostgreSQL [6].

5.1.2 Shortcomings

Conflict resolution for offline sync scenarios where multiple devices edit the same flashcard simultaneously is not fully addressed.

5.1.3 Future Works

Planned improvements include a Redis [7] caching layer for performance optimization, improving the sync conflict resolution strategy for multi-device scenarios and adding application-level rate limiting and usage analytics for the AI features.

5.2 Chrome Extension

5.2.1 Conclusion

The Chrome extension (Manifest V3 [4]) was successfully implemented as the learning interface, providing in-page word interaction with popup definitions, CEFR-based vocabulary classification, a video subsystem with subtitle capture and audio recording, an eBook reader, a flashcard creation and editing workspace, settings management, and English and Turkish user interfaces. The extension operates with a service-worker background architecture and communicates with the backend API for data persistence and AI features.

5.2.2 Shortcomings

The tokenizer and lemmatizer, while enhanced with contraction expansion and morphological rules, are rule-based and may misclassify irregular or domain-specific vocabulary. The dictionary.json file must be manually placed in the workspace directory. Subtitle capture depends on platform-specific DOM structures (e.g., YouTube) which may break when those sites update their layout. The video audio capture via `captureStream()` is not available on DRM-protected content.

5.2.3 Future Works

Replacing the rule-based lemmatizer with a more robust NLP model, adding support for additional subtitle sources and streaming platforms, implementing a content recommendation engine that suggests reading and video material matched to the user's proficiency level, adding text-to-speech integration for flashcard pronunciation, and expanding the i18n system to support additional interface languages beyond English and Turkish.

5.3 Mobile Application

5.3.1 Conclusion

The React Native [5] mobile application was successfully implemented as the on-the-go learning interface of Syntagma, providing users with SRS-scheduled flashcard review sessions, a gamified

badge system, and real-time progress tracking through weekly and monthly statistics charts. By integrating push notifications with configurable daily reminders and streak warnings, the app ensures consistent user engagement beyond desktop environments.

5.3.2 Shortcomings

The mobile application does not include an offline mode; all core functionality including flashcard review, vocabulary tracking, and statistics requires an active internet connection, which may interrupt the learning experience in low-connectivity environments.

5.3.3 Future Works

A planned improvement is the introduction of offline review support with background synchronization, allowing users to complete study sessions without an active connection and sync results when connectivity is restored. Expanding the analytics screen with per-collection retention curves and predicted vocabulary mastery dates would further enrich the learning insights available on mobile.

5.4 Whole System

5.4.1 Conclusion

Syntagma as a complete system successfully demonstrates the integration of spaced repetition science, AI-powered language assistance, and multi-platform delivery into a cohesive language learning tool. The three subsystems, backend API, Chrome extension, and mobile application, work together through a unified data model and synchronization layer to provide Turkish-speaking learners with a comprehensive English vocabulary acquisition experience across web browsing, video watching, eBook reading, and on-the-go review.

5.4.2 Shortcomings

The system is currently limited to a single language pair (English→Turkish). Cross-platform synchronization, while functional, does not handle all conflict scenarios.

5.4.3 Future Works

Future development should prioritize multi-language pair support to serve a broader audience, comprehensive end-to-end testing across all subsystems.

References

- [1] Open-Spaced-Repetition, “FSRS4Anki (Free Spaced Repetition Scheduler),” GitHub Repository, 2022. [Online]. Available: <https://github.com/open-spaced-repetition/fsrs4anki>. [Accessed 4 Nov 2025].
- [2] Expertium, “SM-2 Algorithm (SuperMemo-2),” GitHub Pages, 2021. [Online]. Available: <https://expertium.github.io/Algorithm.html>. [Accessed 4 Nov 2025].
- [3] Openrouter, “Openrouter.ai,” [Online]. Available: <https://openrouter.ai/>.
- [4] Google, “Manifest V3,” Chrome Developers, [Online]. Available: <https://developer.chrome.com/docs/extensions/mv3/>. [Accessed 10 April 2026].
- [5] I. Meta Platforms, “React Native: A framework for building native apps using React,” React Native, [Online]. Available: <https://reactnative.dev/>. [Accessed 10 April 2026].
- [6] The PostgreSQL Global Development Group, “PostgreSQL Documentation,” PostgreSQL, [Online]. Available: <https://www.postgresql.org/docs/>. [Accessed 10 April 2026].
- [7] Redis, “Redis,” Github, [Online]. Available: <https://github.com/redis>. [Accessed 15 May 2026].
- [8] S. Contributors, “Spring Boot Reference Documentation,” Spring, [Online]. Available: <https://docs.spring.io/spring-boot/index.html>. [Accessed 10 April 2026].
- [9] Expo, “Expo Documentation,” Expo, [Online]. Available: <https://docs.expo.dev/>. [Accessed 10 April 2026].
- [10] Digital Ocean, “Digital Ocean,” Digital Ocean, [Online]. Available: <https://www.digitalocean.com/>. [Accessed 14 May 2026].